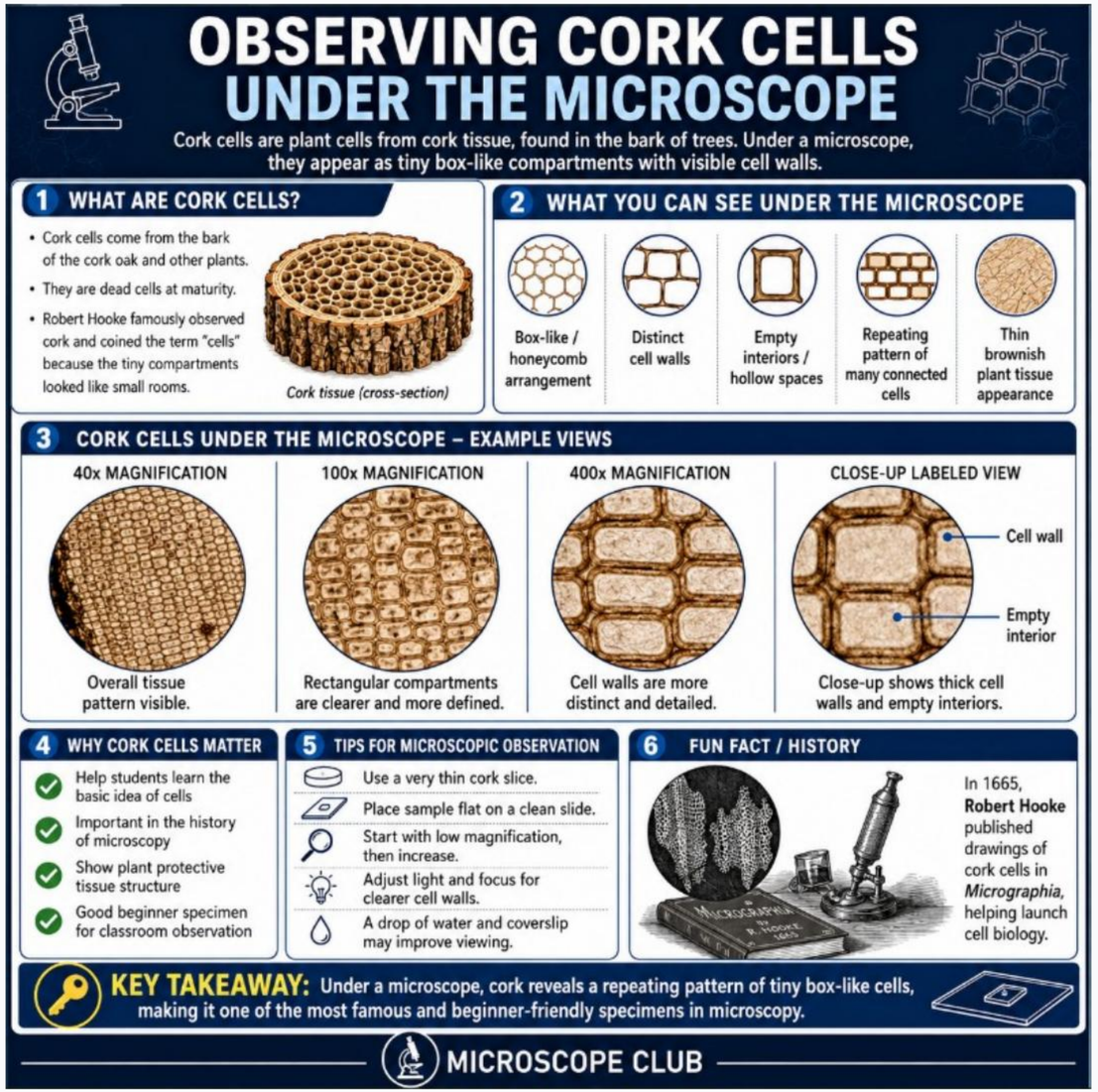
CLASS 8 SCIENCE

# The Fundamental Unit of Life

Discovering cells, basic components, and essential organelles.

# Discovery of the Cell

- **Year 1665:** Robert Hooke's observation.
- **Specimen:** Thin slice of cork.
- **Structure:** Honeycomb-like compartments.
- **Naming:** Coined the term "Cell".



# Three Basic Components



## Cell Membrane

The porous outer boundary.



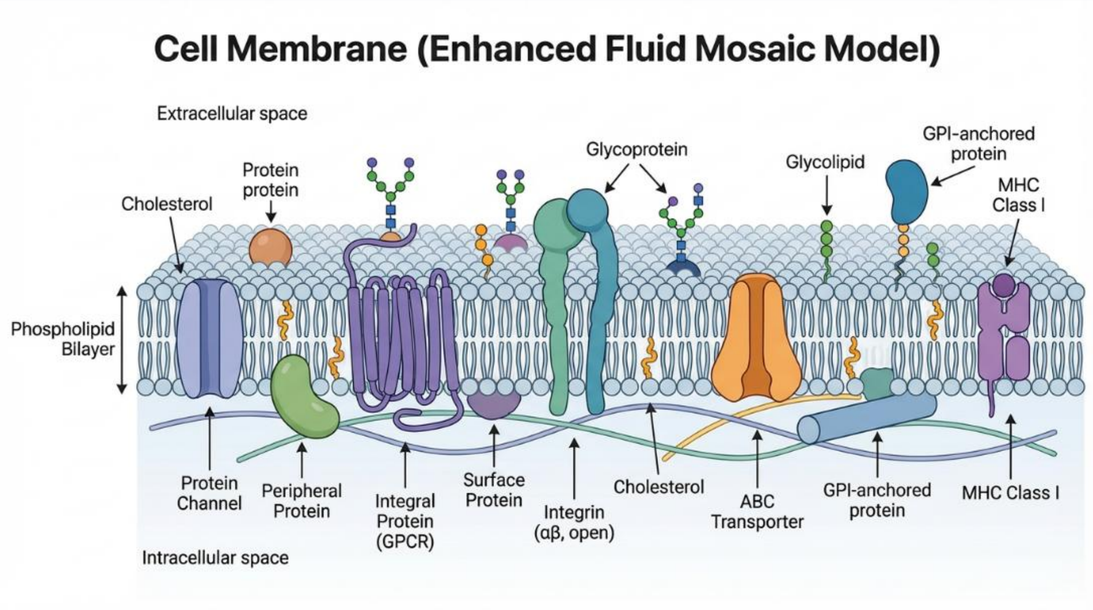
## Cytoplasm

The jelly-like filler.



## Nucleus

The central control unit.




COMPONENT 01

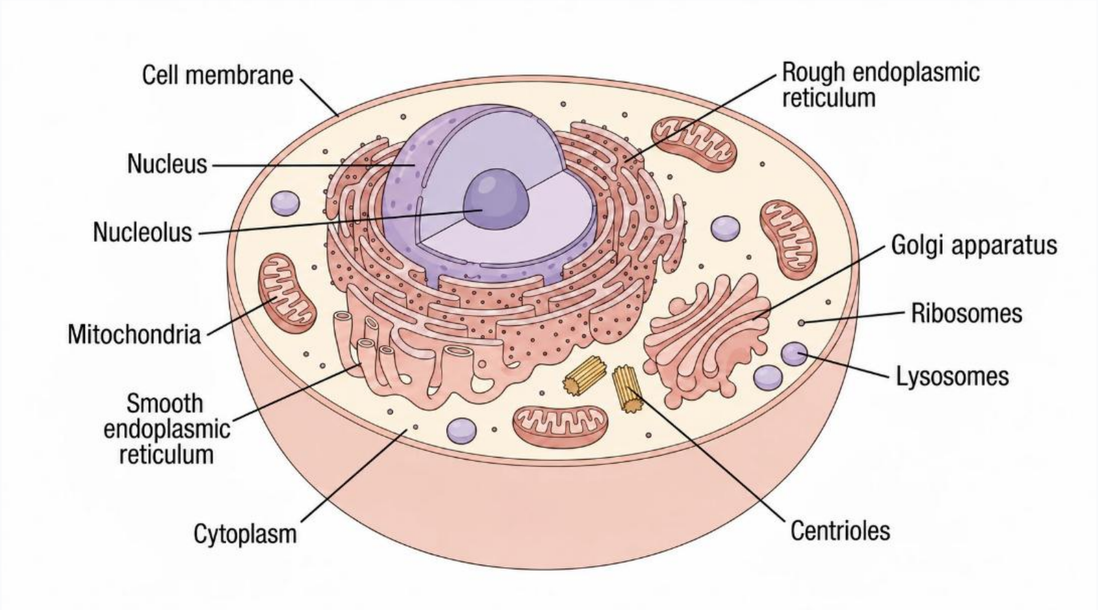
# Cell Membrane

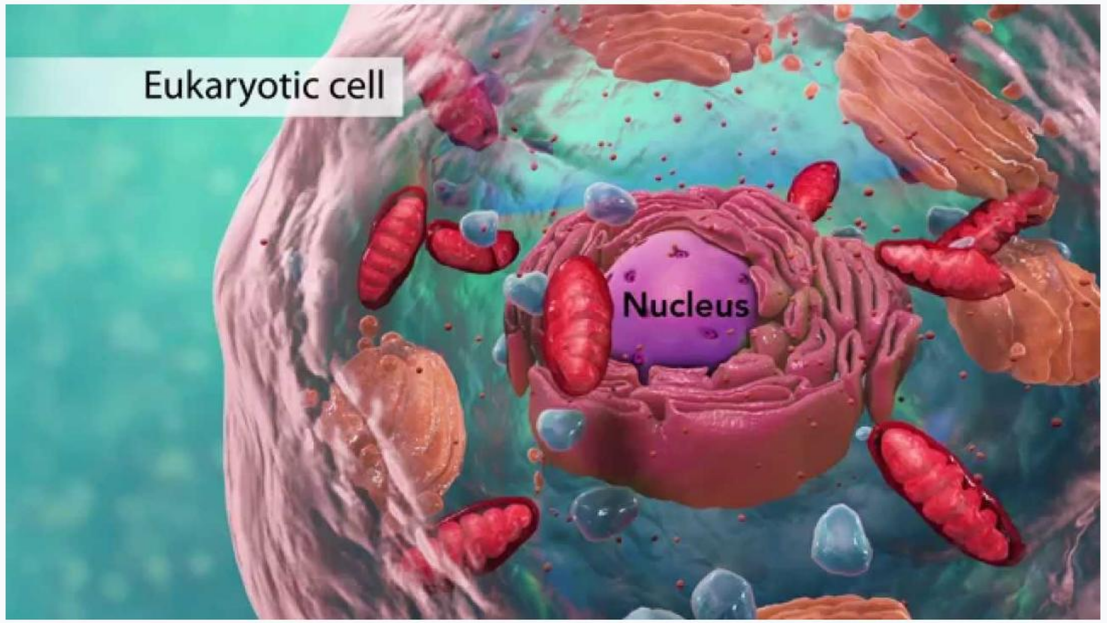
- **Boundary:** Outer covering of the cell.
- **Porous:** Allows specific materials to pass.
- **Gatekeeper:** Regulates entry and exit.

COMPONENT 02

# Cytoplasm

- **Texture:** Thick, jelly-like substance.
- **Location:** Between membrane and nucleus.
- **Suspension:** Holds all cellular organelles.






COMPONENT 03

# The Nucleus

- **Shape:** Dense, spherical central body.
- **Blueprint:** Contains chromosomes and genes.
- **Mastermind:** Controls all cellular activities.

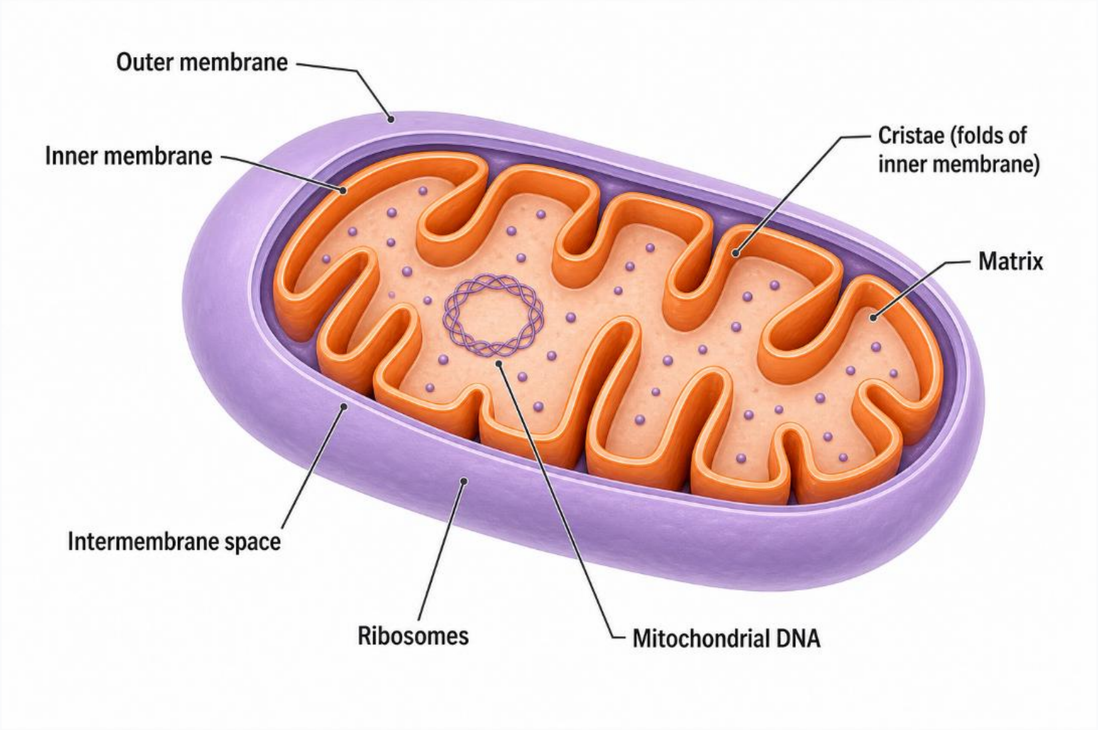
ORGANELLES

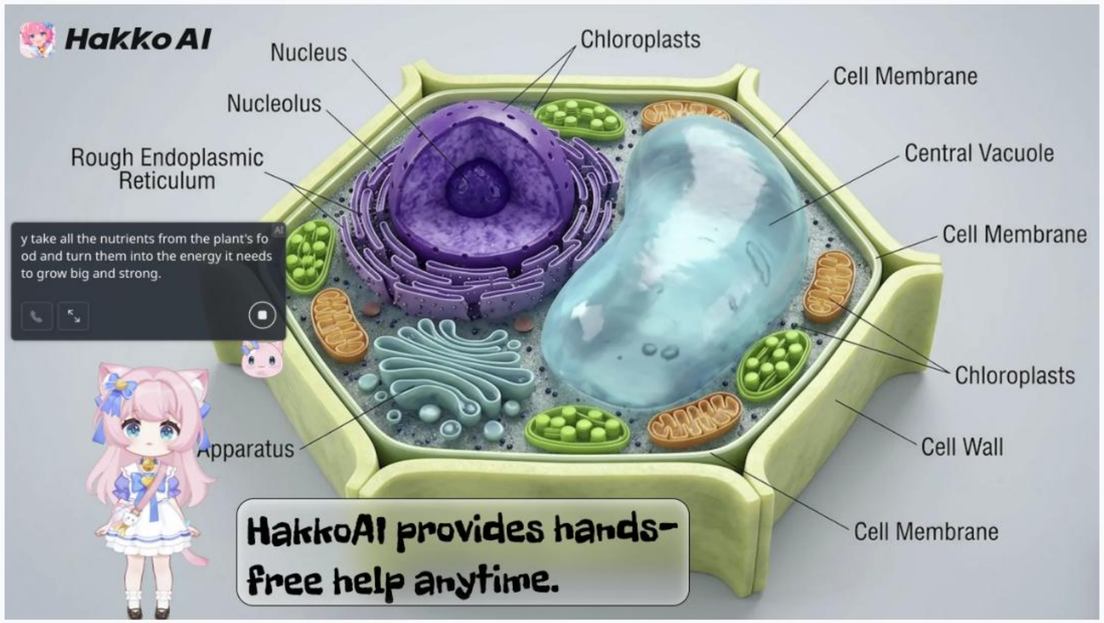
# Mitochondria

**Role:** The powerhouse of the cell.

**Energy:** Produces cellular fuel.

**Function:** Drives cellular respiration.






ORGANELLES

# Vacuoles

- **Function:** Storage sacs for nutrients.
- **Plants:** Feature large, central vacuoles.
- **Animals:** Very small or entirely absent.



# Let's Review

# One-Word Answer Questions

1. **Who discovered the cell in 1665?**
   ✓ Ans: Robert Hooke

2. **Which organelle is called the powerhouse of the cell?**
   ✓ Ans: Mitochondria

# Short Answer Questions

**Q1.** What is the main function of the cell membrane?

**Q2.** Where are chromosomes found inside a cell?

**Q3.** Define cytoplasm in one sentence.

**Q4.** Why do plant cells have exceptionally large vacuoles?

**Q5.** Name the three main structural parts of any cell.

# Long Answer Questions

### Question 1

Explain in detail the discovery of the cell by Robert Hooke.

### Question 2

Describe the structure and function of the cell nucleus.

### Question 3

Differentiate between plant and animal cells based on vacuoles and cell wall.

# Keep Exploring

Biology is the study of life itself.

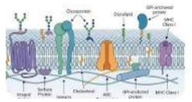
# Image Sources



https://www.microscopeclub.com/wp-content/uploads/2020/12/Observing-Cork-Cells-Under-The-Microscope-1.jpg

Source: www.microscopeclub.com

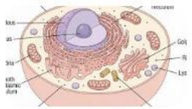
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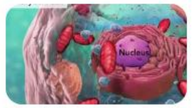
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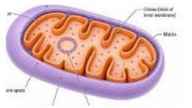
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Source: www.youtube.com

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https://cdn.sci-draw.com/seo/tools/mitochondria-diagram/labeledMitochondria.png

Source: sci-draw.com

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Source: www.tripo3d.ai